

Assignment No-01

Ans to the q: no-01

Transaction errors: Transaction errors are mistakes introduced by the chosen method itself. Think of them as shortcuts that might not be 100% accurate.

Example: Calculating π using formula $\pi = 22/7$ instead of the exact infinite series.

Round of errors: Round of errors happen because computer can not store all digits of a number.

Example: Calculating the square root of 2, because computer can only show limited number of decimal places

Ans to the q. no-02

The root of an equation is necessary because it represents the solution to the equation, indicating the values for which the equation holds true. Identifying roots is essential in solving algebraic problems, as understanding the behavior of functions.

Characteristics of a root:

1. Satisfies the equation: Substituting the root into the equation results in a true statement.

2. Intersection with axis: For a function, a root corresponds to the x -value where the graph intersects the x -axis.

3. Multiplicity: A root can have multiplicity.

4. Complexity: Roots can be real or complex numbers.

5. Dependence on Coefficients: The nature and number of roots depend on the coefficients and degree of polynomial.

Ans to the q: no-03

popular Methods of Finding Roots:

1. Newton-Raphson Method
2. Bisection Method
3. Secant Method
4. False position Method
5. Fixed-Point Method

Bracketing vs. Open Methods

Bracketing Methods: Starts with two guesses that bracket the root. Ex. Bisection, False position.

- Always converge if guesses are correct
- Require opposite signs at endpoints

Open Methods: Start with one or more guesses without needing to bracket the root. Ex. Secant, Newton-Rapson.

- Faster convergence
- Require good initial guesses.

Advantages or Disadvantages

Bracketing Methods

Advantages:

- Guaranteed convergence if bracketing.
- Simple to implement

Disadvantages:

- Slower convergence.
- Requires function evaluations at both ends

Open Methods

Advantages:

- Faster Convergence
- Fewer iterations needed

Disadvantages

- Not guaranteed to converge.
- may require derivatives.

Ans to the q.no-04

Let's calculate the iterations for both the False Position and Second methods to find the root of $f(x) = 5\cos^2 x - e^{2x}$.

Initial Guesses

$$x_0 = 0$$

$$x_1 = 1$$

Function Evaluations

$$f(0) = 5\cos^2(0) - e^0 = 5 - 1 = 4$$

$$f(1) = 5\cos^2(1) - e^1 \approx -5.9301$$

False Position Method iteration

Iteration-1:

$$x_2 = 0 - \frac{4(1-0)}{-5.9301-4} \approx 0.403$$

$$f(0.403) = 1.9931$$

Iteration-2:

$$x_2 = 0.403 - \frac{0.664(1-0.403)}{-5.9301-0.664} = 0.5534$$

$$f(0.5534) = 0.5904$$

Iteration - 3

$$x_2 = \frac{0.5534 \cdot (-5.9296) - 1 \times 0.5904}{-5.9296 - 0.5904} = 0.5940$$

$$f(0.5940) = 0.1595$$

Iteration - 4

$$x_2 = \frac{0.5940(-5.9296) - 1 \times 0.1595}{-5.9296 - 0.1595} = 0.6056$$

$$f(0.6056) = 0.0255$$

Iteration - 5

$$x_2 = \frac{0.6050 \times (-5.9296) - 1 \times 0.025}{-5.9296 - 0.0255}$$

$$= 0.6063$$

$$f(0.6063) = 0.0056$$

The approximate root after 5 iteration of the false position method is $x \approx 0.6063$

Secant Method iterations:

$$y'(x_2) = 5 \cos^2(x) - e^x = -5.9296$$

$$f(-5.9296) =$$

$$x_2 = 1 - \frac{-5.9296(1-0)}{-5.9296-4} \approx 0.403$$

Iteration-2:

$$y(x_2) = 5 \cos^2(0.403) - e^{2 \times 0.403} = 1.99$$

$$x_3 = 0.553$$

Iteration-3

$$y(x_3) = 5 \cos^2(0.553) - e^{2 \times 0.553} = 0.595$$

$$x_4 = 0.553 - \frac{0.595(0.553 - 0.403)}{0.595 - 1.999} \approx 0.617$$

Iteration-4:

$$y(x_4) = 5 \cos^2(0.617) - e^{2 \times 0.617} \approx -0.112$$

$$x_5 = 0.617 - \frac{(-0.112) \times (0.617 - 0.553)}{-0.112 - 0.595} = 0.627$$

Iteration-5:

$$y(x_5) = 5 \cos^2(0.627) - e^{2 \times 0.627} \approx 0.225$$

$$x_6 = 0.627 - \frac{(0.225) \times (0.627 - 0.627)}{0.225 - 0.627} \\ \approx 0.627$$

The approximate root after 5 iteration of the second method is $x = 0.627$.

After 5 iterations, both methods converge to the approximate root $x \approx 0.3690$

Ans to the q.no-05

The Bisection Method cannot diverge if applied correctly.

Condition for Convergence:

1. Continuous Function: The function must be continuous on the interval $[a, b]$
2. Sign Change: There must be sign change between $f(a)$ and $f(b)$.

Example: Consider the function

$$f(x) = x^2 - 1$$

Choose the interval

$$f(0) = -1$$

$$f(2) = 3$$

As long as the initial interval has opposite signs at the endpoints, the Bisection method will converge to a root. If the condition are violated, the method simply cannot be applied, not that it diverges.

Ans to the q. no-06

Slope of Equation: The slope of a function $y = f(x)$ represents the rate of change is defined as the derivative

$$\text{slope} = f'(x)$$

Significance of slope in Newton's Method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

for determining the next approximation and ensuring convergence to the root efficiently.

Ans to the q.no-07

Loop - Breaking Conditions in Numerical Root finding:

1. Convergence Criteria:

→ Absolute Error: Stop if

$$|u_{n+1} - u_n| < \epsilon$$

→ Function value: Stop if

$$|f(u_n)| < \epsilon$$

2. Maximum Iterations: Set maximum number of iteration, N . Terminate if:

$$n \geq N$$

3. Relative Error: Use relative error:

$$\left| \frac{x_{n+1} - x_n}{x_{n+1}} \right| < \epsilon$$

4. Divergence Detection: Stops if updates exceed a threshold.

Demonstration Example:

Solving $f(x) = x^2 - 2 = 0$

1. Initial formula:

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n}$$

2. Initial guess: $x_0 = 1.5$ with tolerance $\epsilon = 10^{-6}$

3. Iteration:

$$\rightarrow f(1.5) = 0.25, f'(1.5) = 3$$

$$\text{Apply the formula: } x_1 = 1.5 - \frac{0.25}{3} = 1.4167$$

Continue iterating until $|x_{n+1} - x_n| < 10^{-6}$

The loop breaks when $|x_{n+1} - x_n|$ is less than 10^{-6} .

- A steeper slope leads to faster convergence
- The slope sign determines the direction of the next approximation.

Example:

For the function:

$$f(x) = x^2 - 2$$

1. Initial Guess: $x_0 = 1$

2. Calculate Slope:

$$f'(x) = 2x \Rightarrow f'(1) = 2$$

3. Newton's Method Update:

$$x_1 = 1 - \frac{f(1)}{f'(1)} = 1 - \frac{-1}{2} = 1.5$$

4. Next iteration:

$$f(1.5) = 0.25, f'(1.5) = 3$$

update:

$$x_2 = 1.5 - \frac{0.25}{3} \approx 1.4167$$

The slope is crucial in Newton's Method

The root is approximately 1.414213

Ans to the q: no-08

Trapezoidal Rule of integration:

The trapezoidal Rule approximates the definite integral of a function by dividing the area under the curve into trapezoids.

Formula

For $f(x)$ over $[a, b]$:

$$\int_a^b f(x) dx \approx \frac{b-a}{2} (f(a) + f(b))$$

For n subintervals:

$$\int_a^b f(x) dx \approx \frac{h}{2} \left(f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right)$$

$$\text{when } h = \frac{b-a}{n}$$

Significance of Error:

The error E_T in the Trapezoidal Rule is given by

$$E_T = -\frac{(b-a)^3}{12n^2} f''(\xi)$$

for some $\xi \in [a, b]$.

key points:

- Decreases with n : Error is inversely proportional to n^2
- Depends on $f''(\xi)$: Highly curved functions yield larger errors

Conclusion: Understanding the errors helps assess the accuracy of the approximation and optimize the number of subintervals for desired precision.

Ans to the q: no-09

To find the area under curve $f(x)$
 $= 5x^4 - 3x^3 + 6x^2 - 4$ from $x=0$ to $x=5$:

Step-1: $a=0, b=5$

Step-2: Let, $n=3$

Step-3:

$$h = \frac{b-a}{n} = \frac{5-0}{3} = \frac{5}{3}$$

Step-4:

$$x_0 = 0$$

$$x_1 = 5/3$$

$$x_2 = 10/3$$

$$x_3 = 5$$

Step-5

$$f(0) = -4$$

$$f(5/3) \approx -3.74$$

$$f(10/3) \approx 172.22$$

$$f(5) = 2900$$

Step - 6: Apply Simson $3/8$ Rule

$$\begin{aligned}\int_0^5 f(x) dx &\approx \frac{3h}{8} (f(0) + 3f(5/3) + 3f(10/3) + f(5)) \\&= \frac{3 \cdot \frac{5}{3}}{8} (-4 + 3 \times 3.74 + 3 \times 172.22 + 2900) \\&= \frac{5}{8} (-4 + 11.22 + 516.66 + 2900) \\&= 2140.86\end{aligned}$$

So, the area under the curve from $x=0$ to $x=5$ is approximately 2140.86.

Ans to the q. no-10

Interpolation: Interpolation is the process of estimating unknown values that fall within the range of known data points.

Example: Given points $(1, 2)$, $(2, 3)$ and $(3, 5)$ using the linear interpolation to estimate

the value at $n = 2.5$ would give a value that lies exactly on the line connecting $(2, 3)$ and $(3, 5)$.

Approximation: Approximation is the process of finding a simple function that closely matches a set of data points, often used for values outside the range of known data points.

Example: To fit the curve to the points $(1, 2)$, $(2, 3)$, and $(3, 5)$, ^{we} you can use polynomial or least square methods. For instance $f(n) = n + 1$ is a linear approximation that the general trend but doesn't pass all points exactly.